
AI Multimodal Flood Disaster Response DataLake (FDR) India

Reference Report

Where the data comes from · How it is collected & displayed · Dashboard working & features · The role of AI

Version 1.0.0

23 September 2026

763 district monitoring zones · 36 states/UTs · 80 CWC river gauges · 7 live real-time feed families · 8
AI-engine modules

FDR — AI Flood Dashboard (India)

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1. Project Idea & How the Dashboard Works

FDR is a full-stack, real-time **flood monitoring and emergency-response dashboard for all of India**. It continuously collects weather, river, satellite and official-alert data, uses an AI engine to score flood risk for every one of India's **763 districts** (0–100 severity), and displays the result on a single screen.

The dashboard is one screen. A full-India Leaflet map sits underneath with two floating side panels:

- **Left — "Flood Situation · All India":** AI plain-language brief, live river water levels, district rainfall, official alerts, data-source health, timeline.
- **Right — "Emergency & Evacuation":** helplines & agencies, safe route planner, evacuation zones, shelters, NDRF/SDRF teams, and the citizen self-service (check my area / report flooding).

How it stays live (working cycle):

1. **Collect** — the backend pulls every live feed in parallel every **60 s**.
2. **Score** — the AI engine scores all zones and builds the state snapshot.
3. **Store** — data goes into a JSON DataLake + a SQLite history lake (7-day time series powering the sparklines).
4. **Show** — the full snapshot is pushed to the browser over a **WebSocket** (`/api/ws`); if the socket drops, the browser reconnects in 3 s and polls the panel endpoints every **20 s** as a fallback.
5. **First load** — done in **one** request (`/api/initial`), so all panels paint immediately instead of waiting on many slow per-feed calls.

Stack: React + Vite + Leaflet (frontend) · FastAPI + asyncio (backend) · JSON + SQLite storage.

2. Data Sources & How Each Is Collected

2.1 Live international feeds (polled every 60 s, no key needed)

Source	What it provides	How it is collected
Open-Meteo Weather	Rainfall, temperature, humidity, wind + 3-day hourly forecast for every district	WeatherClient batches all ~763 zones into multi-point /forecast requests (100 coordinates per call, 6 concurrent, fail-soft per batch) on every refresh cycle.
Copernicus GloFAS (via Open-Meteo Flood API)	River discharge for all 80 CWC gauges — current vs. historical median (flow anomaly) + 10-day flow forecast	FloodForecastClient sends multi-point /flood requests (100 stations per call); each gauge's reading is enriched into the water-level panel.
GDACS (JRC/UN-OCHA)	Verified international flood alerts	OfficialAlertsClient polls the flood event list (180-day look-back), then filters events to India's bounding box.
NASA EONET	Flood events tracked from space	Same alert client polls the floods category (365-day window), filtered to India.

Source	What it provides	How it is collected
RainViewer	Live rain-radar animation on the map	map_tiles fetches the radar index (cached 5 min) and builds the radar overlay layers.
NASA GIBS	Satellite precipitation (IMERG) + daily true-colour imagery	Tile URLs are regenerated once per day; served straight to the map.
Nominatim (OpenStreetMap)	Place search / geocoding for the search box	GET /api/places/search queries Nominatim with India-bounded results.

2.2 Indian government sources

These registrations on the source board are **honest**: a portal that cannot be machine-polled is shown as a *reference link*, and a key-gated API is shown as

needs key — never as fake-live.

Source	What it provides	How it is collected
IMD (India Meteorological Department)	Official district forecasts, basin QPF, state forecasts	IMDClient calls the IMD API when IMD_API_KEY is set; without a key the same rainfall model IMD uses is mirrored from Open-Meteo and the source is marked <i>needs key</i> .
CWC Flood Forecasting (FFS)	Official flood forecasts for 360 stations / 20 basins	Reference portal; its published warning & danger levels per gauge are built directly into the platform, and live river flow comes from GloFAS.
CWC / NWDP telemetry	Real measured gauge water levels	When NWDP_API_KEY is set, NWDPConnector polls river_level_telemetry per station (Bearer auth) and its readings replace the modelled level for that gauge.
ISRO Bhuvan FEWS	Satellite-based flood inundation maps	Reference portal for satellite flood maps.
ISRO Bhoonidhi (Sentinel-1 SAR)	Real flood-extent polygons from radar imagery	Credential-gated download; the client lists the Sentinel-1A SAR collection, and without BH00NIDHI_API_KEY the source shows <i>needs key</i> (no fake polygons in real-time mode).
NDMA SACHET	Official CAP-format disaster alerts	CAP alerts ingested programmatically when SACHET_API_KEY is available; otherwise shown as reference/needs-key.
NDEM · India-WRIS	National disaster portal · water-resources portal	Reference portals (institutional login); NDEM contributes the official flood-inundation WMS layer definition.

2.3 Platform-owned sources

Source	What it provides	How it is collected
Citizen reports	Crowd-sourced flood reports (text + optional photo, English/Hindi, GPS)	Anyone submits via the dashboard (POST /api/citizen/reports). The AI parses it into a structured incident, a FDR-xxxxx ticket is issued, and it feeds the incident map + zone score.
Official bulletin uploads	CWC/IMD/NDMA/state-EOC bulletins in CSV/JSON/GeoJSON	EOC staff upload via POST /api/official/ingest; CWC rows become live gauge readings and verified incidents.

Source	What it provides	How it is collected
Demo scenario (simulation)	11-step monsoon-surge timeline	A scenario player (SIMULATION_ENABLED, off in production) escalates river levels, SAR extents, alerts and scores for guided demos.

Health tracking: every client reports back to a central Source Registry, so the dashboard's **Data-source board** always shows each source's true status (LIVE / DEGRADED / NEEDS_KEY / REFERENCE), latency, record count and last-success time — including which feeds are genuinely streaming right now.

3. Role of AI in the Project

The "AI" is a **transparent decision-support engine**, not a black-box model. There are no trained ML models and no LLM calls in the code — everything is rules, math and classic text parsing, so **every result can be explained and audited**. Eight modules work in sequence on each 60-second cycle:

3.1 Flood risk scoring — the core AI

Every district gets a **0–100 severity score** from a weighted combination of five signals:

$\text{Score} = 0.35 \times \text{Satellite} + 0.20 \times \text{Rain} + 0.20 \times \text{River} + 0.15 \times \text{Reports} + 0.10 \times \text{Population}$

- **Satellite:** flooded-area share and water depth from SAR extents.
- **Rain:** current and forecast rainfall intensity (+ wind).
- **River:** how close gauge water level is to warning/danger levels.
- **Reports:** number and severity of citizen/alerts incidents.
- **Population:** exposed population and vulnerable share.

Result bands: LOW -> MODERATE -> HIGH -> VERY HIGH -> CRITICAL. A national

regional score combines all zones, and each zone tracks a trend so "deteriorating" areas are flagged early.

3.2 Explaining the score

Every zone's score is decomposed — **how many points each of the five factors contributed** — and turned into a plain sentence, shown on the map and via the API. Example: *"CWC gauge on Brahmaputra shows water level 49.2 m vs danger 49.6 m — contributes 33 points."*

3.3 Detecting conflicting data

Cross-checks the sensors for contradictions (e.g. **satellite says flooded but no rain; heavy rain but river stable**) and flags them with a likely cause, so operators know when a source may be stale.

3.4 Prioritising rescue & response

Zones are ranked **CRITICAL -> HIGH -> MONITOR -> LOW** using severity, population exposure, vulnerable population, accessibility and rate of change — with concrete recommended actions (evacuate riverbanks, deploy NDRF/SDRF, stage medical teams).

3.5 Safe route risk

Any origin -> destination is scored **SAFE / CAUTION / UNSAFE** by flood exposure along the route, known flood-cutoff roads and incident density, and a safer detour is recommended.

3.6 Decision recommendations

A five-tier action ladder (tactical deployment, medical staging, evacuation, early warning, data-quality checks) — each recommending the **responsible agency** (e.g. NDRF 9th Battalion) and estimated resources.

3.7 Understanding citizen reports (NLP)

Free-text reports are parsed without any ML library — pure pattern-based NLP — into: **location** (district + place name), **severity** (0.3–0.95), **category** (flooded road, trapped people, power outage, bridge collapse, etc.), **coordinates** and a **confidence score**. This instantly converts crowd input into structured, map-ready incidents.

3.8 The plain-language brief

All the numbers are translated into an everyday summary — "**What this means for you**": risk level with a traffic-light colour, headline, what to do, people at risk, worst-affected areas and an honest note on which feeds are live.

Reference notes prepared for the FDR project.